Induced Subgraphs in Ramsey Graphs

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Abstract

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1 Introduction

Theorem 1.1. *If G is a graph on n vertices with less than dn^2 edges, then G contains a homogeneous set of size greater than $\frac{c}{d\log(1/d)}\log n$.*

Corollary 1.2. *For any C-Ramsey graph, there exists $\delta = \delta(C) \in (0, 1/2)$ such that the edge density of G lies in the interval $[\delta, 1-\delta]$.*

Proof. Apply Theorem 1.1 to G and its complement. □

2 Subgraphs with Many Degrees

In this chapter, we present the following theorem of Bukh and Sudakov.

Theorem 2.1 (Bukh–Sudakov [1])**.** *Every Ramsey graph on n vertices induces a subgraph on $\Omega(n)$ vertices with $\Omega(\sqrt{n})$ distinct degrees.*

Not only is this result interesting on its own right, but its proof also introduces a crucial characteristic of Ramsey graphs:

Definition 2.2. An n-vertex graph G is (c, δ)-*diverse* if for $x \in V(G)$, we have $|N(x)\Delta N(y)| < cn$ for at most n^δ vertices $y \in V(G)$.

The notion of diversity captures the idea that the neighbourhoods of most pairs of vertices differ significantly, which serves as a cornerstone of the argument in the following chapter. The proof of Theorem 2.1 proceeds by showing that every Ramsey graph contains a diverse subgraph of linear size, followed by a probabilistic argument that diversity furnishes many distinct degrees.

Lemma 2.3. *For $\delta \in (0, 1)$, there exists $c = c(C, \delta) > 0$ such that every C-Ramsey graph G on n vertices contains a (c, δ)-diverse subgraph on $\Omega(n)$ vertices.*

Proof. Suppose not. Let $K = \lceil \gamma C \log(C + 1) \rceil$ for some sufficiently large constant $\gamma = \gamma(\delta)$, $m = 8^{-K}n$, and $c = 1/K$. We inductively construct a sequence of disjoint vertex sets $S_1, S_2, \ldots, S_K$ of size m^δ, such that for all S_i one of the following conditions holds:

(i) $d(S_i, S_j) < 8c$ for $j > i$

(ii) $d(S_i, S_j) > 1 - 8c$ for $j > i$

Alongside, we also construct a sequence of induced subgraphs $G = G_0 \subset G_1 \subset \cdots \subset G_K$, such that $|G_i| \geq |G_{i-1}|/8$ and $S_i \subseteq V(G_{i-1})$ for all i.

Suppose $G_1, \ldots, G_{i-1}$ and $S_1, \ldots, S_{i-1}$ have been constructed. By induction, $|G_{i-1}| \geq 8^{1-i}|G| \geq m$. Note that G_{i-1} is not (c, δ)-diverse by assumption, so there exist vertex $x \in V(G_{i-1})$ and set $S_i \subseteq G_{i-1}$ of size m^δ such that $|N_{G_{i-1}}(x)\Delta N_{G_{i-1}}(y)| < c|G_{i-1}|$ for $y \in S_i$. Let $T_i = V(G_{i-1}) \setminus S_i$. Since $m^\delta \leq m/2$ for large enough n, we have $|T_i| \geq |G_{i-1}|/2$.

Suppose $|\overline{N}_{T_i}(x)| \geq |T_i|/2$. Then for $y \in S_i$,

$$d(y, \overline{N}_{T_i}(x)) = \frac{|N_{T_i}(y) \cap \overline{N}_{T_i}(x)|}{|N_{T_i}(x)|} < \frac{c|G_{i-1}|}{|T_i|/2} \leq 4c.$$

Partition $\overline{N}_{T_i}(x)$ into $V_i \sqcup W_i$, where $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) < 8c\}$. Notice

$$8c|W_i| \leq \sum_{v \in W_i} d(v, S_i) \leq \frac{e(\overline{N}_{T_i}(x), S_i)}{|S_i|} = \frac{|\overline{N}_{T_i}(x)|}{|S_i|} \sum_{v \in S_i} d(y, \overline{N}_{T_i}(x)) \leq 4c|\overline{N}_{T_i}(x)|,$$

so $|W_i| \leq |\overline{N}_{T_i}(x)|/2$. But then $|V_i| \geq |\overline{N}_{T_i}(x)|/2 \geq |T_i|/4 \geq |G_{i-1}|/8$. Set $G_i = G[V_i]$. Since $d(S_i, U) < 8c$ for all $U \subseteq V_i$, condition (i) holds. In the case that $|N_{T_i}(x)| < |T_i|/2$, setting $V_i = \{v \in \overline{N}_{T_i}(x) : d(v, S_i) > 1 - 8c\}$ yields the analogous result for condition (ii) by the same argument. This completes the induction.

We may assume that $r \geq K/2$ of the sets $S_1, \ldots, S_K$ satisfy condition (i), say $S_{i_1}, \ldots, S_{i_r}$. Put $S = \bigsqcup_{k=1}^{r} S_{i_k}$. Since each S_{i_k} has size m^δ and $c = 1/K$,

$$d(S) \leq \frac{\binom{m^\delta}{2}}{\binom{rm^\delta}{2}} + \max_{(k,l)} d(S_{i_k}, S_{i_l}) \leq \frac{1}{r} + 8c \leq \frac{10}{K} \cdot \frac{1}{r}$$

Thus Theorem 1.1 yields a homogeneous set A in S with $|A| \geq \frac{\nu K}{10 \log(K/10)} \log |S|$, for some $\nu = \nu(C)$. Note that $|A|$ is also a homogeneous set in G. But then

$$\log |S| \geq \frac{1}{\delta} \log m = \frac{1}{\delta} \log n - \frac{K \log 8}{\delta}.$$

Pick $\gamma > \nu K / 10\delta \log(K/10)$ and we have $|A| > C \log n$ for large enough n, contradiction. $\square$

3 Subgraphs of Quadratically Many Sizes

Definition 3.1 ((δ, ε)-rich graph)**.**

Theorem 3.2 (Kwan–Sudakov 2020 [])**.** *Every C-Ramsey graph on n vertices contains induced subgraphs of $\Omega(n^2)$ distinct sizes.*

Proof. □

4 The Erdős–McKay Conjecture

4.1 The Full Erdős–McKay Conjecture and Its Resolution

4.2 Quadratic Polynomials of Gaussians

4.3 The Carbery–Wright Theorem

Theorem 4.1 (Carbery–Wright []).

4.4 Fourier Analysis and Characteristic Functions

4.5 Additive Structure Dichotomy

4.6 The Two-Scale Decomposition

4.7 The Buckets Argument

Theorem 4.2 (Kwan–Sah–Sauermann–Sawhney 2023 []).

5 Conclusing Remarks

References

[1] Boris Bukh and Benny Sudakov. Induced subgraphs of ramsey graphs with many distinct degrees. *Journal of Combinatorial Theory, Series B*, 97(4):612–619, 2007. ISSN 0095-8956. doi: https://doi.org/10.1016/j.jctb.2006.09.006. URL https://www.sciencedirect.com/science/article/pii/S0095895606001080.

A Supplementary Proofs